Predicting Flight Prices for the Singapore Traveller

Leveraging Machine Learning to Beat Volatility and Identify Optimal Booking Windows

Predicting Flight Prices for the Singapore Traveller

Summary

Information Trap

- Current travel platforms only show today's price
- Difficult to estimate the best time to purchase flight tickets

Predictive Solution

- Built a Random Forest machine learning model
- Models 8 major bidirectional routes
- Analyses historical trends and demand signals to forecast upcoming airfares

The Solution – Dynamic Fare Optimiser

- Translates complex price elasticity into simple consumer recommendation
  - Estimates airfare based on purchase and departure date
  - Gives buy/wait guidance and when to purchase tickets

Eliminating Booking Uncertainty

The Volatility Trap

<u>The Problem</u>

- Current travel platforms only show today’s price
- Airline pricing rules are confidential and change constantly
- Travellers cannot easily predict the best time to buy a ticket

<u>The Goals</u>

- Analyse historical flight price data to spot hidden price trends
- Help budget-conscious consumers avoid expensive pricing traps

Shifting Power Back to the Buyer

Using fare intelligence to eliminate booking uncertainty

Our Target Audience

1. Flexible leisure travellers
   - Users seeking value who possess the flexibility to optimise booking dates
2. Budget-conscious consumers
   - Individuals requiring high-precision data to navigate dynamic pricing and secure the absolute price floor

Our Potential Solution: Prescriptive Booking Intelligence

- Automated forecasting
  - Moves beyond simple price tracking to analyse fare patterns
- Actionable guidance
  - Delivers a clear "Buy vs. Wait" signal, transforming raw data into a reliable, time-saving purchasing strategy.

Project Scope and Boundaries

Defining the parameters to capture diverse aviation market dynamics

<u>Flight Routes</u>

- **SIN ↔ Bangkok (BKK)** (Short haul / high frequency)
- **SIN ↔ Tokyo (NRT)** (Corporate / Tourism imbalance)
- **SIN ↔ London (LHR)** (Long haul / geopolitical focus)
- **SIN ↔ Melbourne (MEL)** (Transit feeder hub)

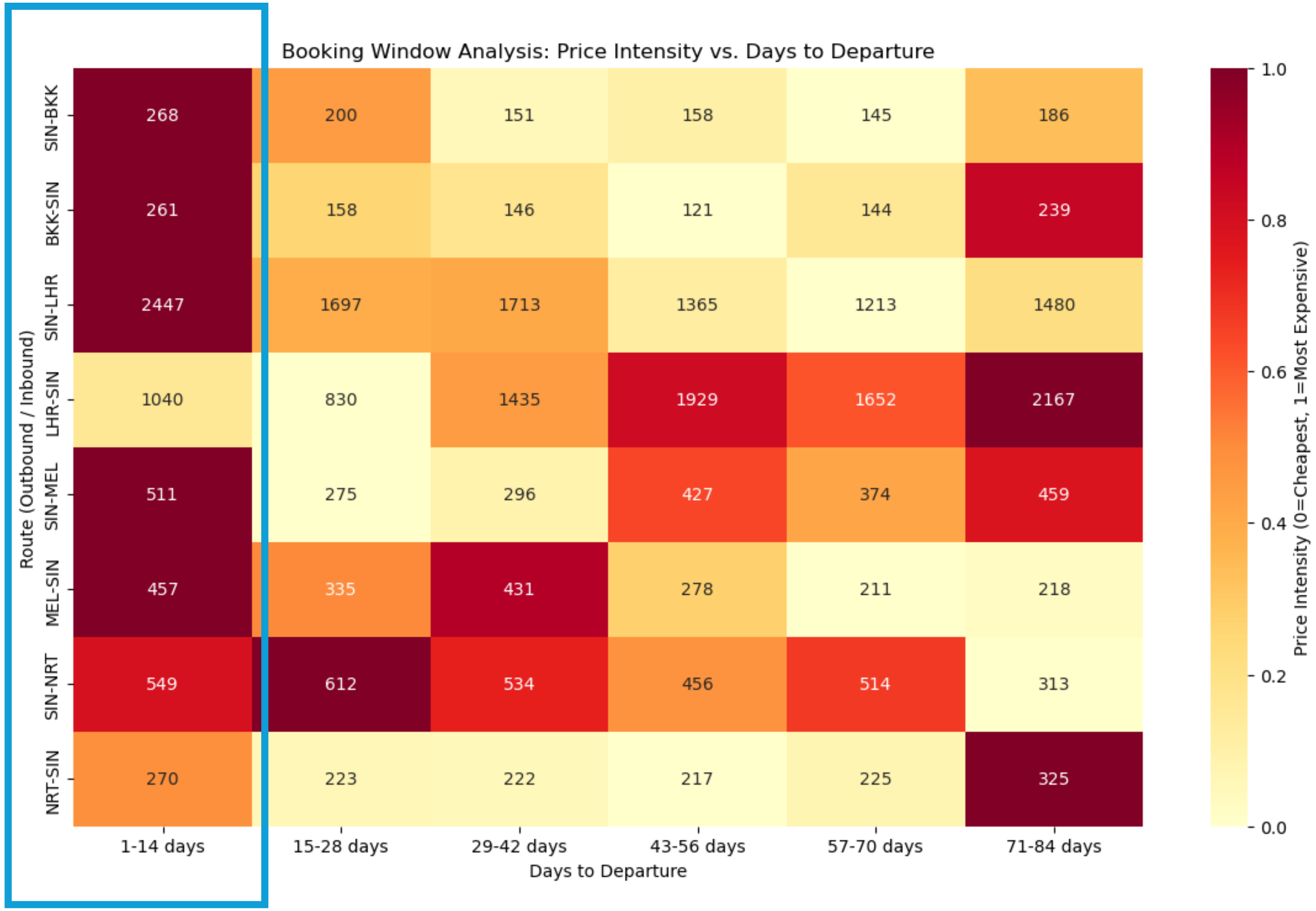
<u>Carrier Types:</u> Low cost carriers (LCC) and full-service carriers (FSC)

<u>Study Parameters:</u>

- **Booking Horizon:** ~84 days out to departure
- **Search Dates Window:** 5 March 2026 to 26 May 2026

Last Minute Volatility Cliff

Airline yield management systems extract premiums from urgent demand

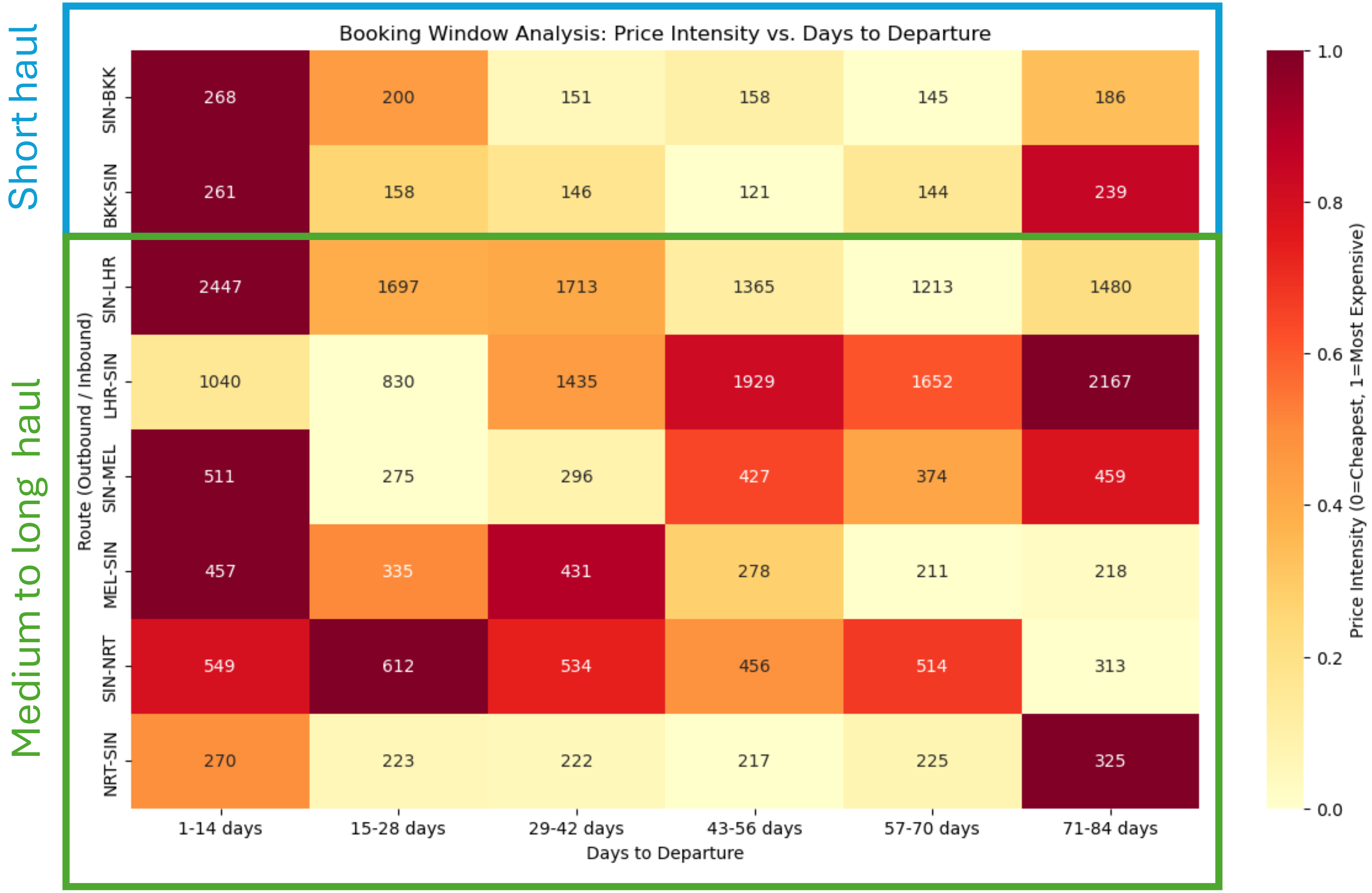


Last Minute Price Escalation

- 1-14 days before departure, fare dispersion vanishes and prices converge to a single, high-intensity floor.
- Profit maximization strategy
  - Airlines systematically unbundle promotional fare classes, forcing last-minute travellers into premium-priced buckets.
- Buyer’s Disadvantage
  - This window represents the lowest information asymmetry for airlines—they know the traveller's need for a seat outweighs their sensitivity to price.

Directional Asymmetry

The inverse demand cycles driving long haul fare variance



Market Maturity (BKK)

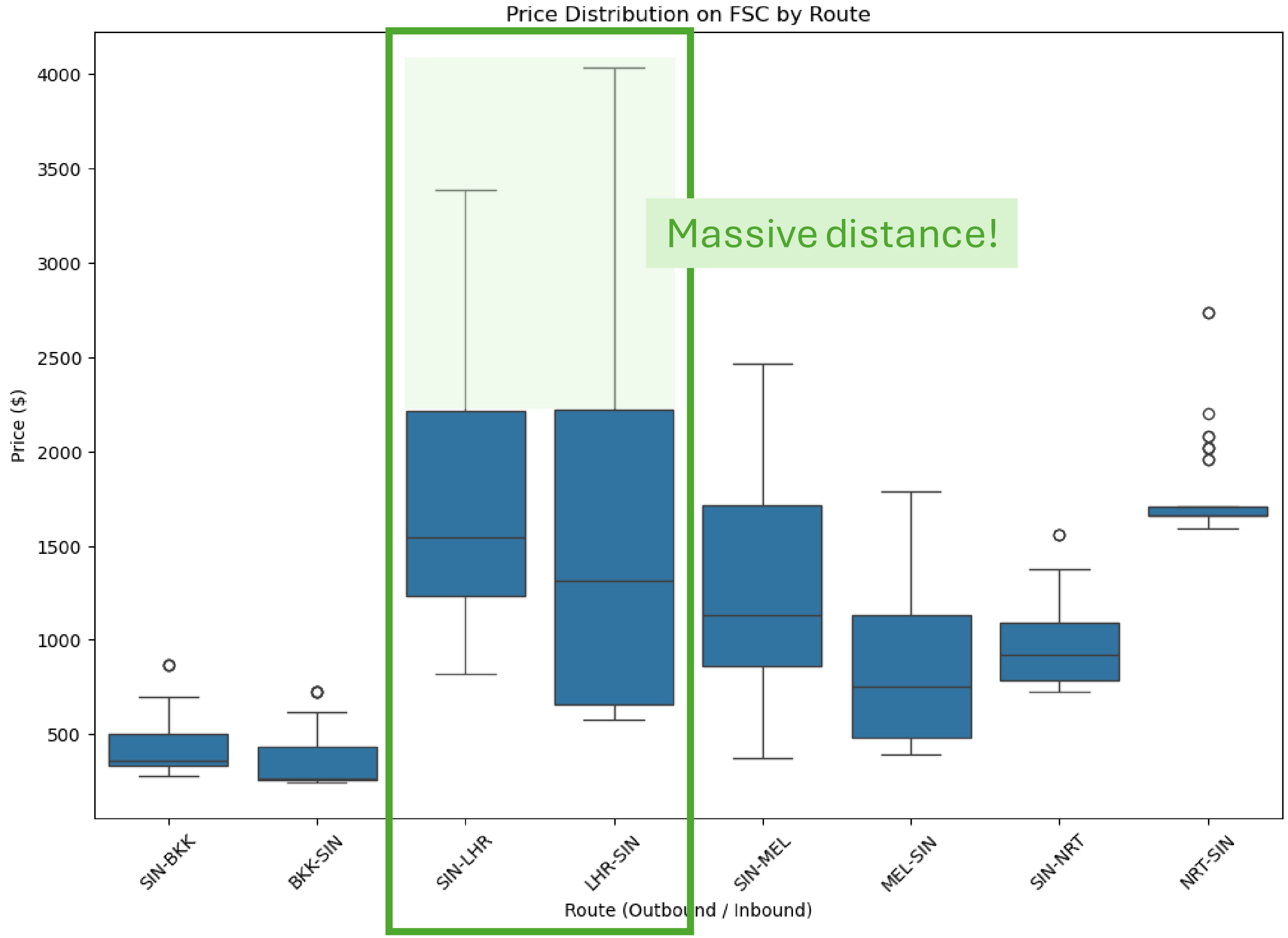
- High frequency and carrier dense
    - Price convergence
    - Low directional volatility

Directional Asymmetry

- Inverse pricing patterns where peak intensity on the outbound leg frequently coincides with a pricing "lull" on the inbound leg.
- Yield management strategy
    - Directional pricing to maximize load factors, treating each leg as an independent market to offset high-cost long-haul operations.

Geopolitical Risk Premiums

Analysing the correlation between airspace instability and fare volatility



<u>Inventory constriction</u>

- Regional conflicts force flight diversions, and flight suspensions

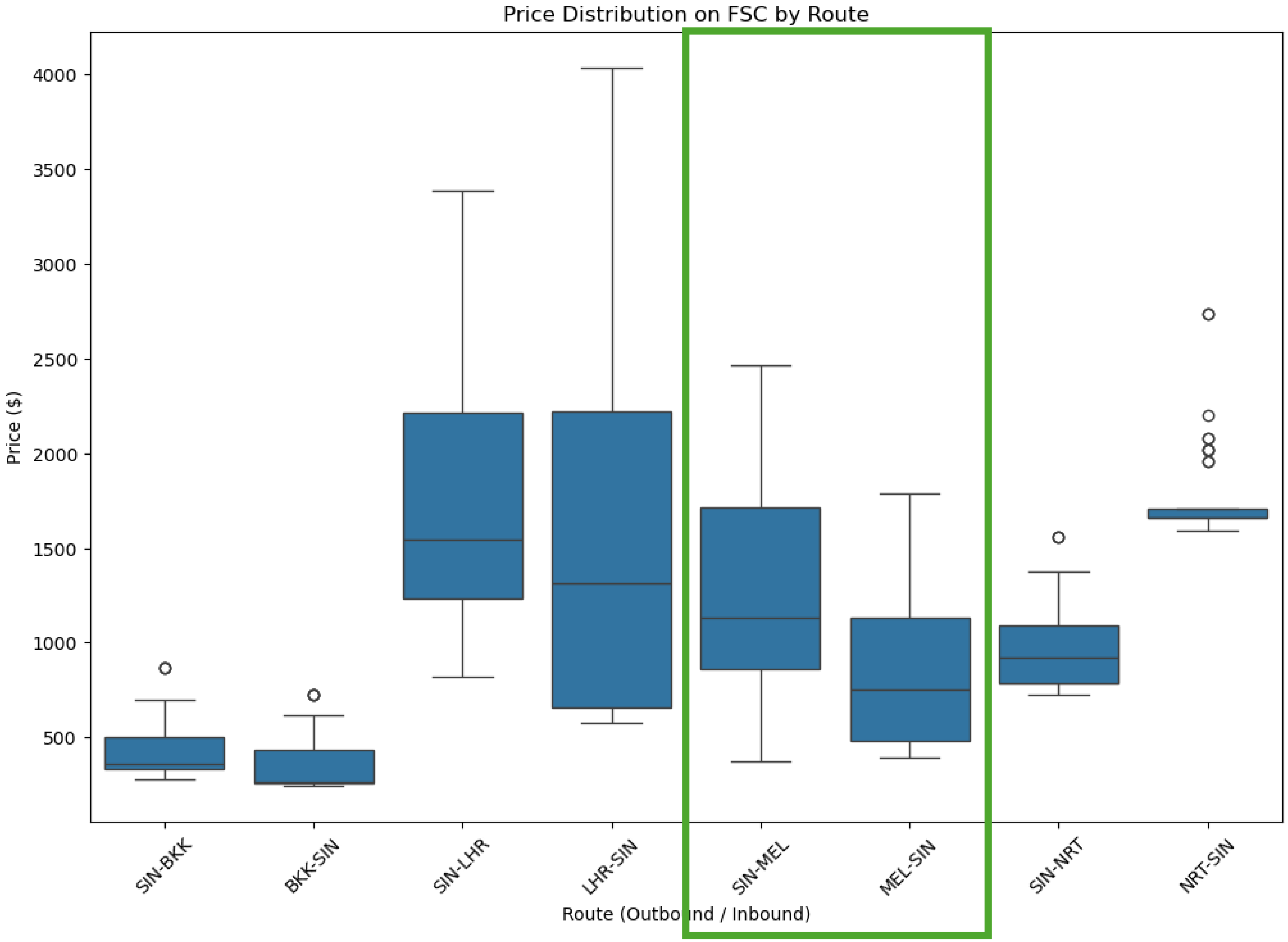
<u>Risk premiums</u>

- High price volatility due to lengthened flight paths, increased fuel consumption, and operational re-routing costs

<u>Geopolitical risk becomes the primary driver of fare fluctuations</u>

Network Effects: Geopolitical Demand Spillovers

Analyzing how regional hub instability creates directional pricing asymmetry



Hub displacement effect

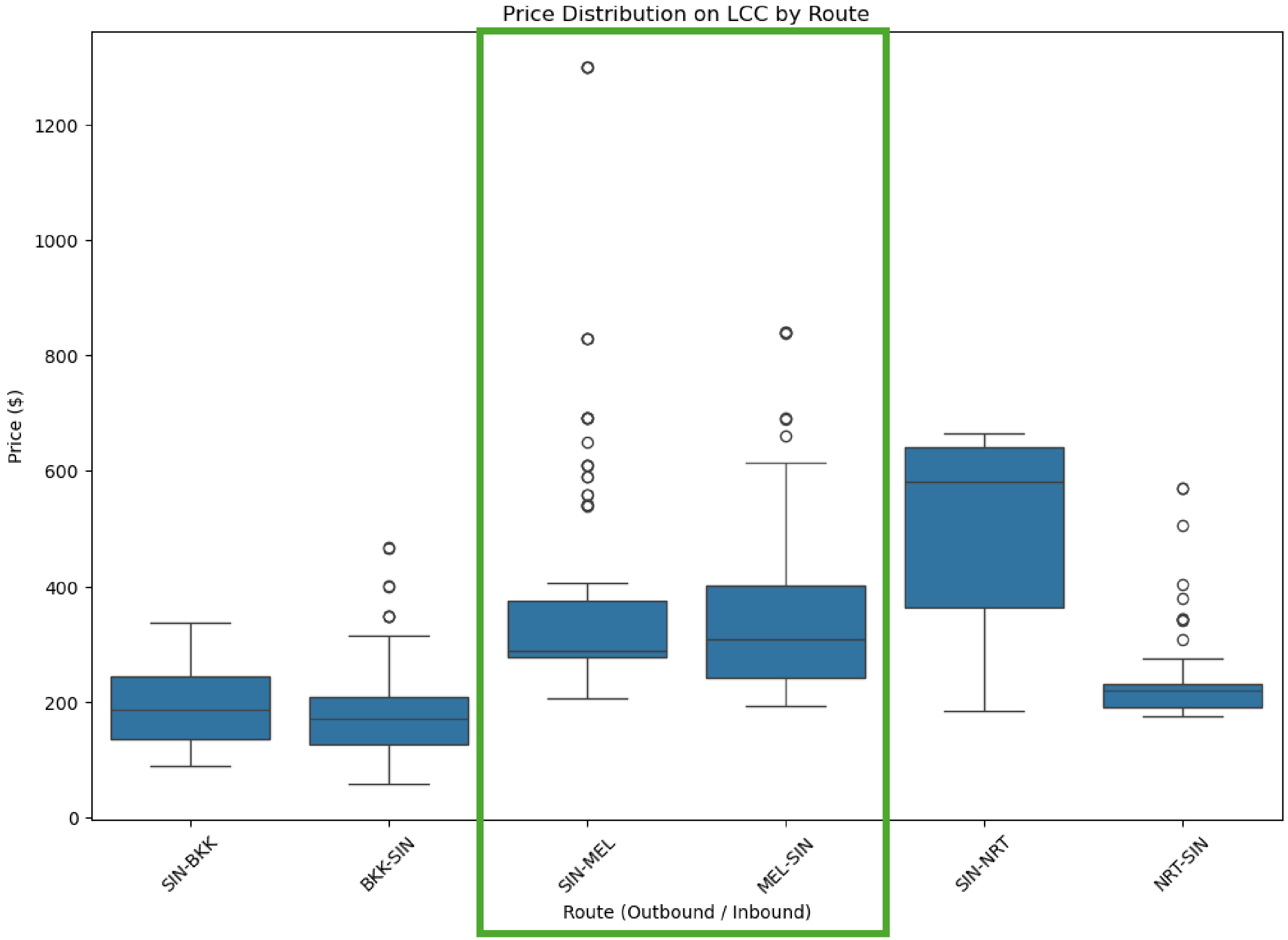
- Geopolitical tensions have forced flight re-routing.
- Singapore Changi emerged as a primary relief hub, absorbing significant passenger displacement.

Directional asymmetry

- SIN-MEL
  - Higher pricing intensity due to the influx of diverted transit passengers competing for limited seat capacity.
- MEL-SIN
  - Operates on a standard return-loop baseline, lacking the same geopolitical demand premium.

Network Connectivity vs. Operational Isolation

Low-Cost Carriers remain insulated from global demand spillover

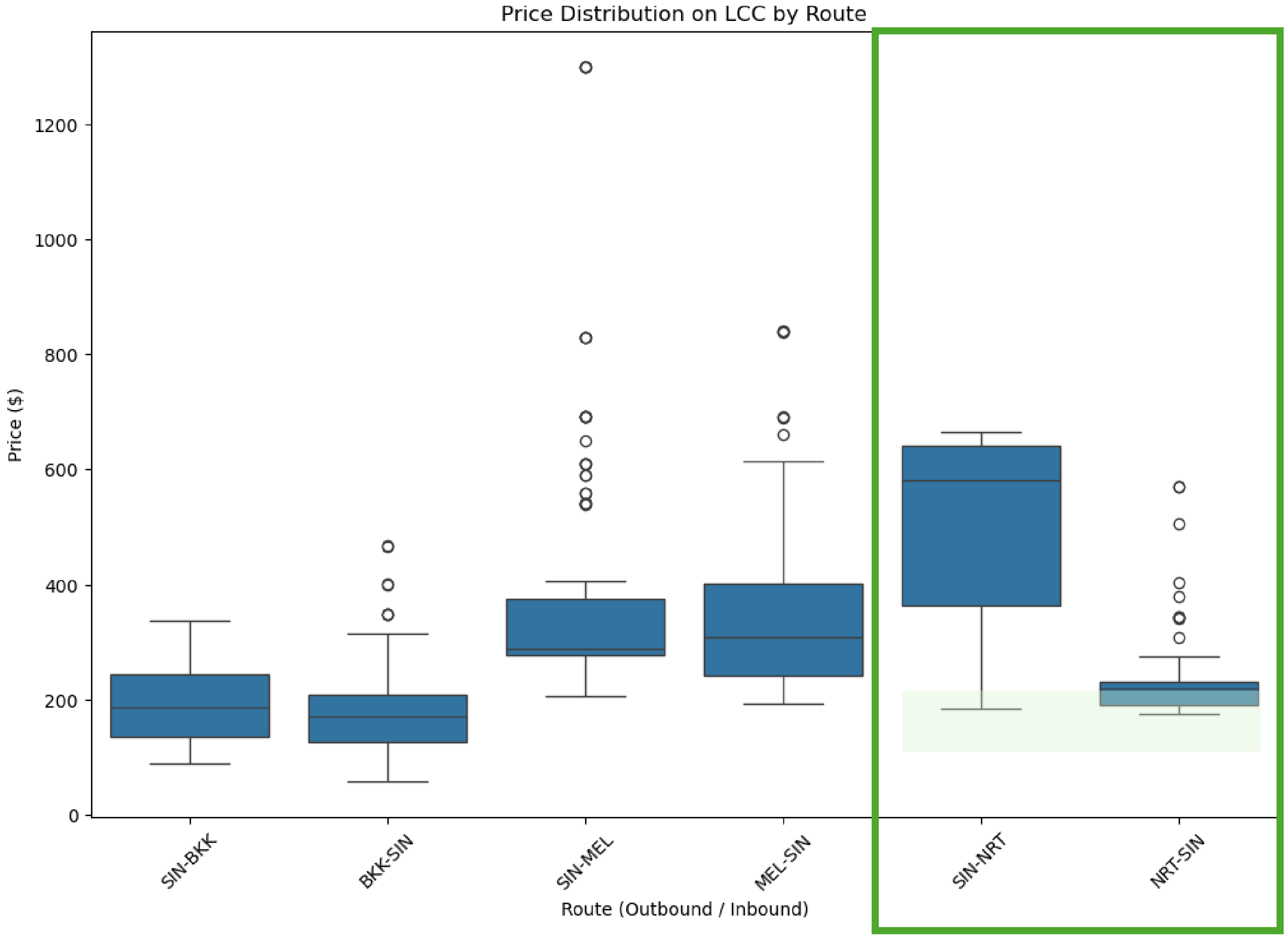


<u>Isolation from Full-Service network dynamics</u>

- Uniform distribution
- Isolated point-to-point systems
  - Interline booking frameworks
  - Alliance partnerships
  - Passenger protection protocols

Decoding the LCC Fare Spectrum

Budget Carriers have directional asymmetry too



Market pricing floor

- Baseline cost remains consistent regardless of direction

Directional asymmetry (NRT)

- SIN-NRT
  - Exhibits a higher median price
  - Indicates robust, consistent demand
- NRT-SIN
  - Displays a lower core price range
  - Premium price outliers, representing opportunistic demand spikes

Capturing Market Demand Drivers

Translating human travel behaviour into predictive machine learning signals

- **School holidays cycles** *(June & December)*
  - Identified peak family travel periods
  - Capture the "fare bucket" shifts driven by family travel demand
- **Travel event windows** *(Public Holidays ± 2 day buffer)*
  - Implemented a buffered event-window logic to account for traveller behavior in bridging public holidays with annual leave for optimised travel
- **Cyclical weekly demand** *(Saturday & Sundays)*
  - Standardised weekly price-cycle logic, capturing the consistent premium placed on weekend departures vs. mid-week troughs

Model Success Metrics

How we validate the accuracy of our price predictions

1. **Core Accuracy (MAPE < 25.0%)**
   - Margin of error – average of percentage errors across all flights
   - Normalises errors – prevents high cost tickets from skewing our understanding of model consistency
     - Prevents a big $500 error on a $2,000 ticket from being treated worse than a $500 error on a $100 ticket
   - Ensures core pricing logic is with high relative consistency
2. **Explanatory Power ($R^2 > 85.0\%$)**
   - Proves that price changes in the market are correctly captured and explained in our model
3. **Prediction Stability (RMSE < $300)**
   - Absolute error magnitude
   - Ensures model misses are within acceptable bounds for a budget-conscious traveller

MAPE: Mean Absolute Percentage Error ; R2: Coefficient of Determination ; RMSE: Root Mean Square Error

Feature Architecture

Engineering the pricing signals for machine learning

Feature	Description	Sample SIN–NRT (Departure: 24 Jun 2026)
Route	8 bidirectional routes (BKK, NRT, LHR, and MEL)	SIN-NRT
Departure Month	Month of departure date	June
Day of Week	Departure day	2 (Wednesday)
Weekend	Flags if the departure date is a weekend	False
Booking Window	Number of days to departure from booking date in 2 week intervals (up to 84 days)	1-14 days
Low Cost Carrier	Flags a low cost carrier	False
Singapore Public Holiday	Flags public holidays within a ±2-day window of the departure date	False
Other Public Holiday		False
Singapore School Holiday	Flags if the departure date falls within June or December	True

Random Forest Performance & Validation

Confirmation of the model's ability to forecast fare dispersion

1. **Core Accuracy (MAPE < 25.0%)**
   - Margin of error – average of percentage errors across all flights
   - Normalises errors – prevents high cost tickets from skewing our understanding of model consistency
   - Ensures core pricing logic is with high relative consistency

Result: 19.3%

2. **Explanatory Power (R^2 > 85.0%)**
   - Proves that price changes in the market are correctly captured and explained in our model

Result: 85.3%

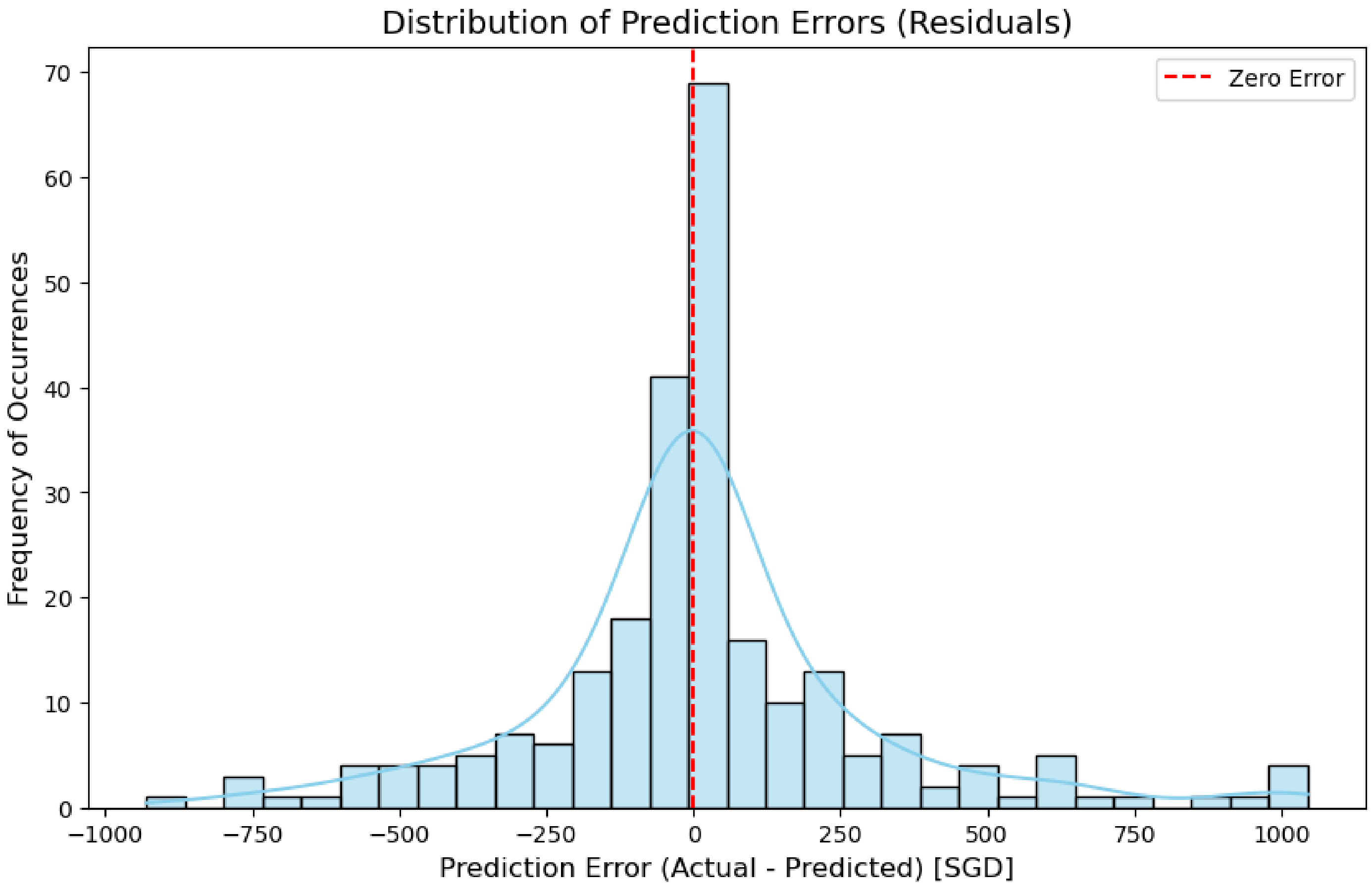
3. **Prediction Stability (RMSE < $300.00)**
   - Absolute error magnitude
   - Ensures model misses are within acceptable bounds

Result: $294.75

MAPE: Mean Absolute Percentage Error ; R2: Coefficient of Determination ; RMSE: Root Mean Square Error

Model Diagnostics: Residual Error Analysis

Validating unbiased performance and characterizing remaining volatility



Unbiased baseline

- Residuals cluster tightly around zero
- Represents high-confidence predictions

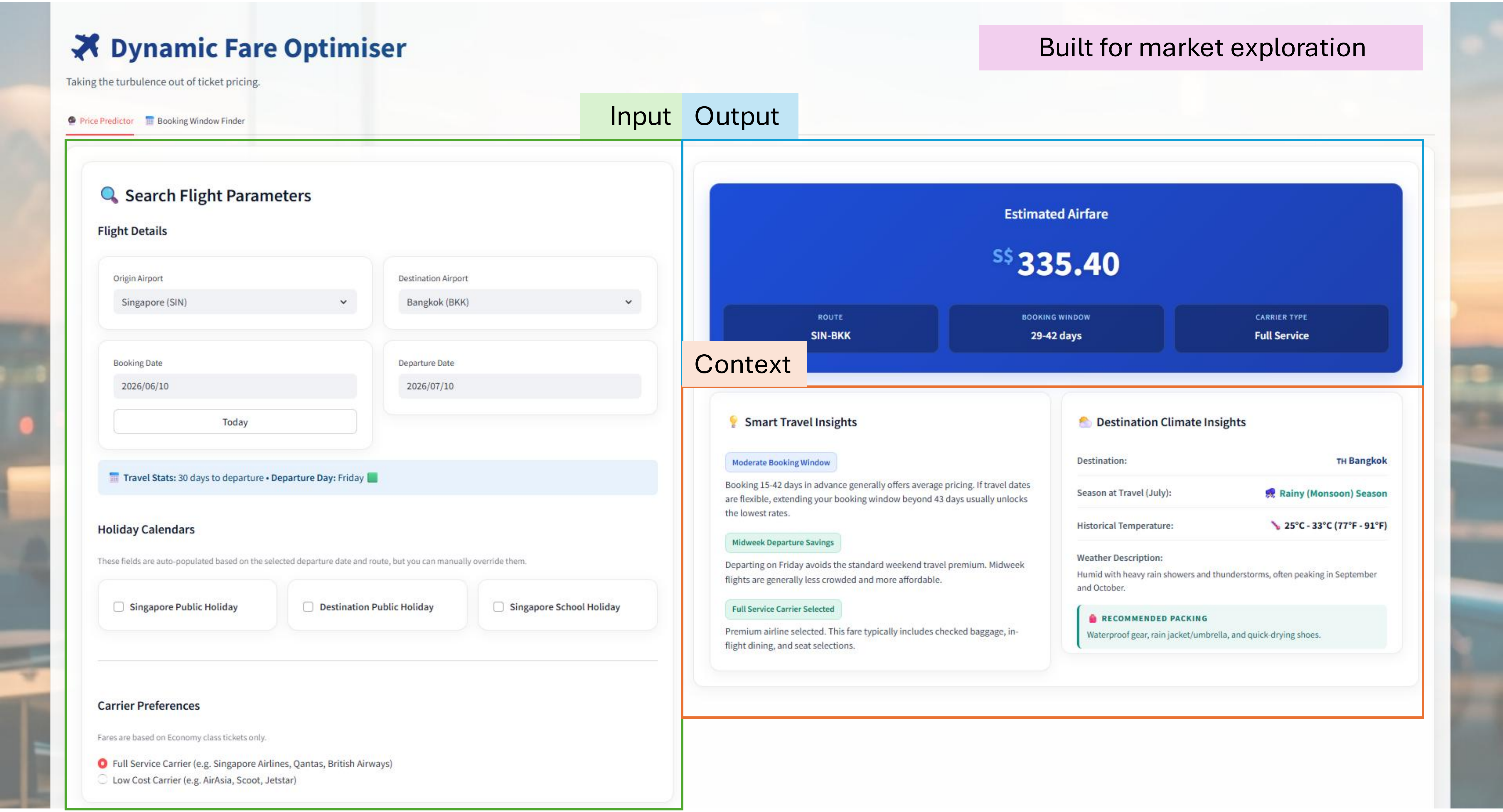
Tails (Variance)

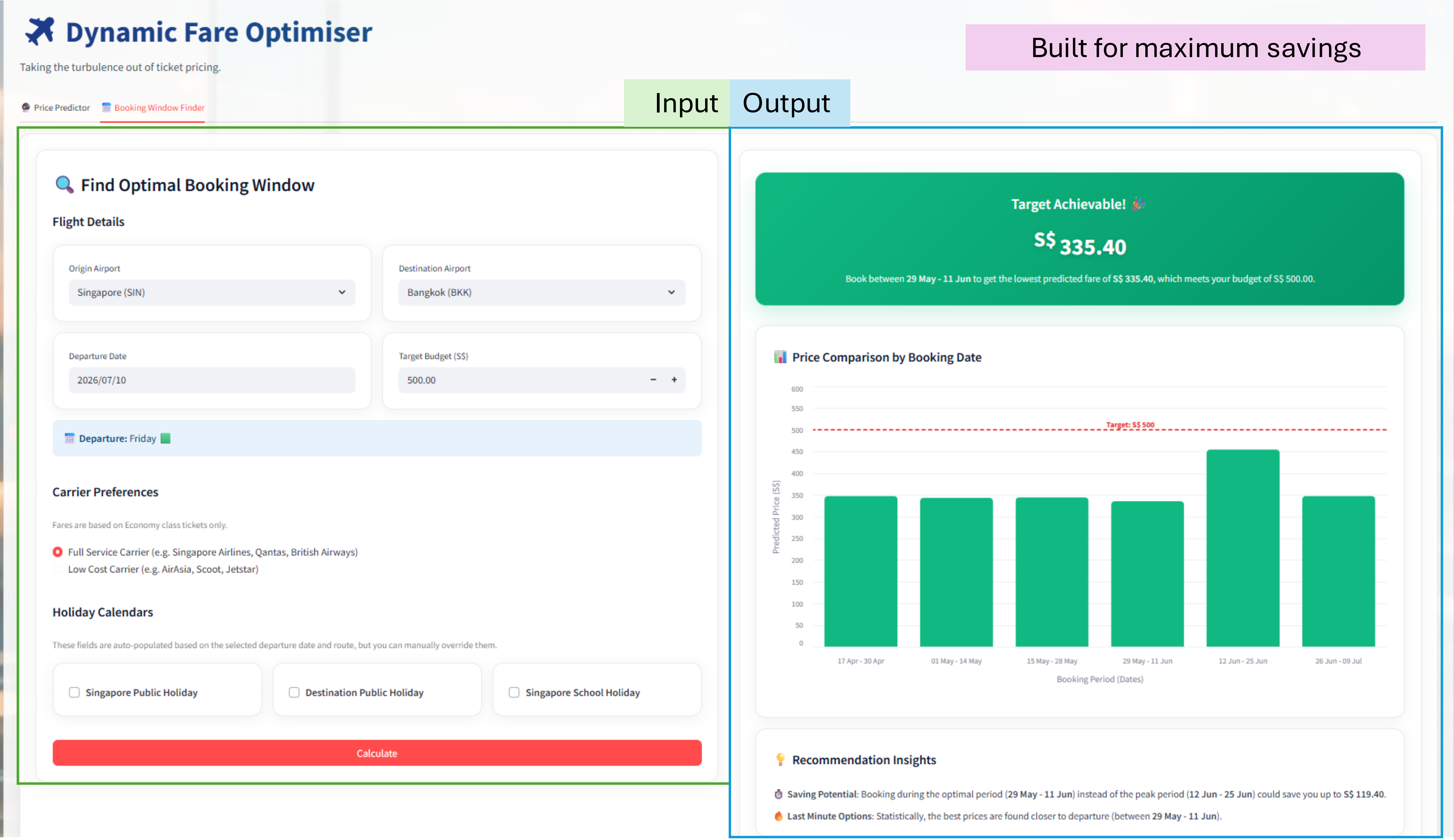
- Represents pricing volatility

Key Pricing Drivers

Carrier type is the most important feature

Feature	Importance (%)
Low cost carrier	44.3
LHR-SIN	9.1
SIN-LHR	7.9
Other Public Holiday	6.8
NRT-SIN	6.0
Departure Month	6.0
Singapore Public Holiday	5.5
SIN-MEL	3.3
SIN-NRT	2.7
Day of Week	1.8


Dynamic Fare Optimiser
Taking the turbulence out of ticket pricing.
Built for market exploration
Price Predictor
Booking Window Finder
Input
Output
Search Flight Parameters
Flight Details
Origin Airport
Singapore (SIN)
Destination Airport
Bangkok (BKK)
Booking Date
2026/06/10
Today
Departure Date
2026/07/10
Travel Stats: 30 days to departure • Departure Day: Friday
Holiday Calendars
These fields are auto-populated based on the selected departure date and route, but you can manually override them.
Singapore Public Holiday
Destination Public Holiday
Singapore School Holiday
Carrier Preferences
Fares are based on Economy class tickets only.
Full Service Carrier (e.g. Singapore Airlines, Qantas, British Airways)
Low Cost Carrier (e.g. AirAsia, Scoot, Jetstar)
Estimated Airfare
S$ 335.40
ROUTE
SIN-BKK
BOOKING WINDOW
29-42 days
CARRIER TYPE
Full Service
Context
Smart Travel Insights
Moderate Booking Window
Booking 15-42 days in advance generally offers average pricing. If travel dates are flexible, extending your booking window beyond 43 days usually unlocks the lowest rates.
Midweek Departure Savings
Departing on Friday avoids the standard weekend travel premium. Midweek flights are generally less crowded and more affordable.
Full Service Carrier Selected
Premium airline selected. This fare typically includes checked baggage, in-flight dining, and seat selections.
Destination Climate Insights
Destination:
TH Bangkok
Season at Travel (July):
Rainy (Monsoon) Season
Historical Temperature:
25°C - 33°C (77°F - 91°F)
Weather Description:
Humid with heavy rain showers and thunderstorms, often peaking in September and October.
RECOMMENDED PACKING
Waterproof gear, rain jacket/umbrella, and quick-drying shoes.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Input
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Price Predictor
Booking Window Finder
Find Optimal Booking Window
Flight Details
Origin Airport
Singapore (SIN)
Destination Airport
Bangkok (BKK)
Departure Date
2026/07/10
Target Budget (S$)
500.00
Departure: Friday
Carrier Preferences
Fares are based on Economy class tickets only.
Full Service Carrier (e.g. Singapore Airlines, Qantas, British Airways)
Low Cost Carrier (e.g. AirAsia, Scoot, Jetstar)
Holiday Calendars
These fields are auto-populated based on the selected departure date and route, but you can manually override them.
Singapore Public Holiday
Destination Public Holiday
Singapore School Holiday
Calculate
Target Achievable!
S$ 335.40
Book between 29 May - 11 Jun to get the lowest predicted fare of S$ 335.40, which meets your budget of S$ 500.00.
Price Comparison by Booking Date
Target: S$ 500
Predicted Price (S$)
600
550
500
450
400
350
300
250
200
150
100
50
0
17 Apr - 30 Apr
01 May - 14 May
15 May - 28 May
29 May - 11 Jun
12 Jun - 25 Jun
26 Jun - 09 Jul
Booking Period (Dates)
Recommendation Insights
Saving Potential: Booking during the optimal period (29 May - 11 Jun) instead of the peak period (12 Jun - 25 Jun) could save you up to S$ 119.40.
Last Minute Options: Statistically, the best prices are found closer to departure (between 29 May - 11 Jun).

Constraints and Strategic Road Map

Refining model architecture to ensure long-term predictive reliability

Type	Limitation	Strategic Scaling Goal
Data Scope	Current model is trained with limited route pairings.	Scale to ingest and learn from a wider diversity of routes.
Automated Pipeline	Current forecasting relies on a static batch snapshot of market prices.	• Implement cloud-native ingestion functions for real-time tracking • Implement monthly model re-training.
Feature Complexity	Public holiday “Travel Event Window” feature implementation contained redundant conditional logic.	Refine holiday demand features to align more precisely with observed market behavior and demand signals.

Model Robustness: Forward Test

Evaluating prediction stability against live market data drift

Target	Measure	Development *(Cut-off 26 May 2026)*	Forward Test *(480 records)*	Change
Core Accuracy	MAPE	19.3%	24.0%	+4.7% points
Explanatory Power	R2	85.3%	74.5%	-10.8% points
Prediction Stability	RMSE	$294.75	$383.42	+$88.67

Data Degradation

- Dynamic pricing schedules drift constantly
- Evolving airline pricing algorithms outpace historical training trends

Strategic Roadmap

- Justifies a 1-Month Model Re-training Pipeline, ensuring the predictive engine continuously re-aligns with shifting global market volatility

Taking Turbulence out of Ticket Pricing

Takeaways

<u>The Dynamic Fare Optimiser</u>

- The random forest model exceeded target benchmarks
  - Core Accuracy (MAPE): 19.3%
  - Explanatory Power (R^2): 85.3%
  - Prediction Stability (RMSE): $294.75
- Carrier type is the dominant pricing signal
- Post 26-May 2026 forward testing indicates a natural degradation in explanatory power
- To mitigate degradation and price volatility, the application should transition into a cloud-native, automated re-training pipeline to ensure model evolution alongside airfare price algorithms.